

11. Prove that $\omega + \omega = \omega 2$.

Solution

Let $A = \{a_1, a_2, \dots\}$, $B = \{b_1, b_2, \dots\}$, $C = \{c_1, c_2, \dots\}$, $D = \{r, s\}$ be well-ordered set.

$$\therefore \omega = \text{ord}(A) = \text{ord}(B) = \text{ord}(C), \text{ord}(D) = 2$$

$$\begin{aligned} \omega + \omega &= \text{ord}(\{A; B\}) \\ &= \text{ord}(\{a_1, a_2, \dots; b_1, b_2, \dots\}) \end{aligned}$$

$$\begin{aligned} \omega 2 &= \text{ord}(\{C \times D\}) \\ &= \text{ord}(\{(c_1, r), (c_2, r), \dots; (c_1, s), (c_2, s), \dots\}) \end{aligned}$$

Let $f: \{A; B\} \rightarrow \{C \times D\}$ be defined by

$$f(x) = \begin{cases} (c_i, r), & x = a_i \\ (c_i, s), & x = b_i \end{cases}$$

Since $\{A; B\}, \{C \times D\}$ be arranged by using $\preceq$,

$\therefore f$ is one-one, onto and preserved order.

$$\therefore \{A; B\} = \{C \times D\}$$

$$\therefore \text{ord}\{A; B\} = \text{ord}\{C \times D\}$$

$$\omega + \omega = \omega 2.$$

12. If A is well-ordered and B is similar to A then B is well-ordered.

Solution Let $(A, \preceq_A)$ be well-ordered $(B, \preceq_B)$ be similar to A .

Then $\exists$ a bijection from A to B : $a_1, a_2 \in A$,

$$a_1 \preceq_A a_2 \Leftrightarrow f(a_1) \preceq_B f(a_2) \quad \dots\dots\dots(1)$$

Let $B' \subset B$ and $B' \neq \emptyset$.

Then $f^{-1}(B') \subset A, f^{-1}(B') \neq \emptyset$.

So $\exists$ 1st element $a_0 \in f^{-1}(B') : a_0 \preceq_A a, \forall a \in f^{-1}(B') \dots\dots\dots(2)$

$$\therefore f(a_0) \in B' + 9\backslash$$

Let $b \in B', \exists a^* \in A : f(a^*) = b$

$$\therefore a^* \in f^{-1}(B')$$

By (2) $a_0 \preceq_A a^*$

By (1) $f(a_0) \preceq_B f(a^*) = b, \forall b \in B'$.

$\therefore B'$ contains 1st element $f(a_0)$.

$\therefore B$ is well-ordered.

7. Let A and B be similar well-ordered sets. Prove that there exists only one similarity mapping of A into B.

Solution

Let $f: A \rightarrow B$ and $g: A \rightarrow B$ be similarity mappings.

Suppose $f \neq g$

$$\exists x \in A: f(x) \neq g(x)$$

$\therefore$ either $f(x) < g(x)$ or $g(x) < f(x)$.

Let $f(x) < g(x)$

Since $g: A \rightarrow B$ be similarity mapping

$g^{-1}: B \rightarrow A$ be similarity mapping

$\therefore g^{-1} \circ f: A \rightarrow A$ be similarity mapping

$$\text{But } f(x) < g(x) \Rightarrow (g^{-1} \circ f)(x) < (g^{-1} \circ g)(x) = x$$

$\therefore (g^{-1} \circ f)(x) < x$ contradict to $x \preceq (g^{-1} \circ f)(x)$, for every $x \in A$.

$$\therefore f = g.$$

$\therefore \exists$ one similarity mapping of A into B.

8. State whether each of the following statements about ordinals is true or false; if it is true prove it and if it is false, give a counter example:

(i) If $\lambda \neq 0$, then $\mu < \lambda + \mu$. (ii) If $\lambda \neq 0$, then $\mu < \mu + \lambda$.

Solution

(i) Let $\mu = \text{ord}(\{1, 2, 3, \dots\})$, $\lambda = \text{ord}(\{a_1, a_2, \dots, a_\lambda\})$

$$\lambda + \mu = \text{ord}(\{a_1, a_2, \dots, a_\lambda; 1, 2, 3, \dots\}) = \mu$$

$\therefore \mu < \lambda + \mu$ is false.

(ii) let $\mu = \text{ord}(A)$, $\lambda = \text{ord}(B) \neq 0$

$$\mu + \lambda = \text{ord}(\{A; B\})$$

Since $B \subset \{A; B\}$, B has 1st element t_0 .

$\therefore A$ is equivalent to $s(t_0)$

$\therefore A$ is shorter than $\{A; B\}$

$$\therefore \mu < \mu + \lambda.$$

9. Prove, by giving a counter example, that the right distributive law of multiplication over addition (for the ordinal numbers) is not true in general. In other words, for three ordinal numbers λ, μ and η

$$(\lambda + \mu)\eta = \lambda\eta + \mu\eta.$$

Solution

$$\text{Let } \lambda + \mu = 1, \eta = \omega$$

$$\text{Let } 2 = \text{ord}(\{a, b\}), \omega = \text{ord}(\{1, 2, 3, \dots\})$$

$$2\omega = \text{ord}(\{(a, 1), (b, 1), (a, 2), (b, 2), \dots\}) = \omega$$

$$\omega 2 = \text{ord}(\{(1, a), (2, a), (3, a), \dots; (1, b), (2, b), (3, b), \dots\}) > \omega$$

$$(\lambda + \mu)\eta = (1 + 1)\omega = 2\omega = \omega.$$

$$\lambda\eta + \mu\eta = 1 \cdot \omega + 1 \cdot \omega = \omega + \omega = \omega \cdot 1 + \omega \cdot 1 = \omega(1 + 1) = 2\omega > \omega.$$

$$\therefore (\lambda + \mu)\eta \neq \lambda\eta + \mu\eta.$$

10. Let A and B be well-order set and let

$$S = \{x \mid x \in A, s(x) = s(y), y \in B\} \text{ and}$$

$$T = \{y \mid y \in B, s(y) = s(x), x \in A\}. \text{ Prove that } S \text{ is similar to } T.$$

Solution

let $x \in S$. $s(x)$ is similar to a unique initial segment $s(y)$ of B.

$$\therefore \forall x \in S, \exists \text{ uniquely } y \in T: s(x) = s(y).$$

Let $f: S \rightarrow T$ be defined by $f(x) = y$ if $s(x) = s(y)$.

Let $f(x) = f(x')$, $x, x' \in S$

$$y = y', y, y' \in T$$

$$\therefore s(x) = s(y) = s(y') = s(x')$$

$$\therefore s(x) = s(x')$$

$$\therefore s(x) = s(x'), x = x'$$

$\therefore f$ is one-one.

Let $y \in T$. Choose $x \in A: s(x) = s(y)$. $\exists x \in S: f(x) = y$.

$\therefore f$ is onto.

Let $x', x \in S$, $f(x) = y$, $f(x') = y'$.

Let $x' < x$

Let $\phi: s(x) \rightarrow s(y) = s(f(x))$ be similarity mapping.

ϕ restricted to $s(x')$ into initial segment $s(\phi(x'))$ of B.

$\phi(x') = f(x') = y'$. Since $\phi(x') \in s(y)$, $\phi(x') < y$

$$\therefore y' < y$$

$$\therefore f(x') < f(x)$$

$\therefore S$ is similar to T .

4. Prove: Let A and B be well-ordered sets and let $S = \{x \mid x \in A, s(x) = s(y), y \in B\}$. Prove that $S = A$ or S is an initial segment of A .

Solution

Let $x \in S$ and $y \prec x$.

$s(y)$ is similar to an initial segment of B ; hence $y \in S$.

Suppose $S \neq A$. Since $S \subset A, A - S \neq \emptyset, (A - S) \cap S = \emptyset$.

$\therefore (A - S) \subset A$

$\therefore (A - S)$ has a first element $a_0, a_0 \notin S$

$s(a_0) = \{x \mid x \in A, x \prec a_0\}$

$x \in s(a_0) \Rightarrow x \prec a_0$

$\Rightarrow x \in (A - S)$

$\Rightarrow x \in S$

$\therefore s(a_0) \subset S$

If $y \in S, a_0 \prec y \Rightarrow a_0 \in S$ which contradicts $a_0 \notin S$

$\therefore y \notin S$

$\therefore S \subset s(a_0)$

$\therefore S = s(a_0)$

Therefore, $S = A$ or S is an initial segment of A .

5. Let A and B be well-ordered sets. Prove that A is shorter than B , A is similar to B or A is longer than B .

Solution

Let $S = \{x \mid x \in A, s(x) = s(y), y \in B\}$

$T = \{y \mid y \in B, s(y) = s(x), x \in A\}$ and $S \sim T$.

There are 4 possibilities.

(i) $S = A$ and $T = B$. Then A is similar to B .

(ii) $S = A$ and $T = s(b)$, initial segment of B . Then A is shorter than B .

(iii) $T = B$ and $S = s(a)$, initial segment of A . Then A is longer than B .

(iv) $S = s(a)$ and $T = s(b)$. Then $s(a) = s(b)$.

$\therefore a \in S$

But $a \notin s(a)$

$\therefore$ (iv) is impossible.

6. Prove the principle of Transfinite Induction.

Solution

Suppose $S \neq A$

Let $A - S \neq \emptyset, (A = S \cup T, S \cap T = \emptyset)$

$\therefore T \subset A$, well-ordered set.

$\therefore$ first element $t_0 \in T$.

$s(t_0) = \{x \mid x \in A, x \prec t_0\}$

$\therefore x \in s(t_0), x \notin T$.

$\therefore x \in S$

$\therefore s(t_0) \subset S$

By (2) $t_0 \in S$

This contradicts that $t_0 \in T$.

$\therefore S \neq A$ is not true.

$\therefore S = A$.

Definition

Let A be an ordered set with the property that every subset of A contains a first element. Then A is called a well-ordered set.

အစု A သည် ordered set ဖြစ်၍ အစုပိုင်းတွင် first element ရှိပါကလည်း

အစု A ကို Well-ordered set ဟုခေါ်သည်။

ဗလာအစုကို Well-ordered set ဟု ယူဆမည်။

Initial Segments

Let A be a well-ordered set. The initial segment $s(a)$ is defined by $s(a) = \{x / x \in A, x < a\}$ where $s(a) \subset S$.

A သည် well ordered set ဖြစ်ပါစေ initial segment determined by a ဆိုသည်မှာ အစုပိုင်း $s(a) = \{x / x \in A, x < a\}$ ဖြစ်၍ $s(a)$ ဖြင့် သင်္ကေတပြုသည်။

1. Let A be well-ordered sets and let S be a subset of A with the property that $a \preceq b$ and $b \in S$ implies $a \in S$. Prove that $S = A$ or S is an initial segment of A .

Solution

Suppose $S \neq A$. Since $S \subset A, A - S \neq \emptyset, (A - S) \cap S = \emptyset$

$$\therefore (A - S) \subset A$$

$$\therefore (A - S) \text{ has a first element } a_0, a_0 \notin S$$

$$s(a_0) = \{x / x \in A, x < a_0\}$$

$$x \in s(a_0) \Rightarrow x < a_0$$

$$\Rightarrow x \notin (A - S)$$

$$\Rightarrow x \in S$$

$$\therefore s(a_0) \subset S$$

If $y \in S, a_0 \preceq y \Rightarrow a_0 \in S$ which contradicts $a_0 \notin S$

$$\therefore y \notin S$$

$$\therefore S \subset s(a_0)$$

$$\therefore S = s(a_0)$$



2. Prove that two different initial segments of a well-ordered set cannot be similar.

Solution

Let A be a well-ordered set.

Let $s(a)$ and $s(b)$ be two different initial segment.

$$\therefore a \neq b$$

$$\therefore \text{either } a < b \text{ or } b < a$$

$$\text{Let } a < b$$

$$x \in s(a) \Rightarrow x < a$$

$$\Rightarrow x < b$$

$$\Rightarrow x \in s(b)$$

$$\therefore s(a) \subset s(b)$$

Initial segment $s(a) \subset s(b)$ well-ordered set.

$\therefore s(b)$ is not similar to $s(a)$.

3. Let A and B be well-ordered sets and let an initial segment $s(a)$ of A be similar to an initial segment of B . Prove that $s(a)$ is similar to a unique initial segment $s(b)$ of B .

Solution

Let A and B be well-ordered sets.

Let $s(a) = s(b)$ and $s(a) = s(b')$, $b, b' \in B, b \neq b'$

$$\therefore s(b) = s(b')$$

two different initial segments of a well-ordered set cannot be similar

$$\therefore s(b) = s(b')$$

$$b < b' \Rightarrow b \in s(b'), b \notin s(b)$$

$$\Rightarrow s(b') \neq s(b)$$

$$b' < b \Rightarrow b' \in s(b), b' \notin s(b')$$

$$\Rightarrow s(b) \neq s(b')$$

$$b = b'$$

$\therefore s(a)$ is similar to a unique initial segment $s(b)$ of B .